MURSHID R

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AI/ML Research Engineer • Applied AI & LLM Systems • Space Technology AI

PROFESSIONAL SUMMARY

B.Tech Computer Science (Data Science & AI) student at Dr. M.G.R Educational and Research Institute with hands-on research experience in deep learning for aerospace applications, LLM-powered systems, and real-time AI pipelines. Built SENTINEL — an end-to-end LLM commitment extraction platform using Groq (Llama 3.3 70B), FastAPI, and React 19, submitted to AMD Slingshot Hackathon 2026. Developed a Temporal Convolutional Network (TCN) achieving 92% accuracy in combustion instability prediction for hybrid rocket engines, presented at YASSC 2025 (State Level). Proficient in RAG architectures, MCP-style agentic systems, LLM prompting, PyTorch, TensorFlow, and full-stack AI deployment. Seeking internships in Applied AI, ML Engineering, AI Architecture, or Space Technology AI.

TECHNICAL SKILLS

Programming: Python (Expert) · TypeScript · JavaScript · C++ · Java · C · SQL · HTML/CSS

AI / ML Frameworks: PyTorch · TensorFlow · Keras · Scikit-learn · HuggingFace Transformers · LangChain

LLM & AI Architecture: RAG (Retrieval-Augmented Generation) · MCP (Model Context Protocol) · LLM Prompting & Fine-tuning · Groq API · Llama 3.3 70B · Google Gemini · Agentic Systems · LLM Inference Pipelines

ML Architectures: Temporal Convolutional Networks (TCN) · LSTM · CNN · Bayesian Optimization · Collaborative Filtering · Sentiment Analysis · Risk Scoring Engines

Backend & APIs: FastAPI · Flask · WebSockets · Celery · REST API Design · JWT Authentication · OAuth 2.0 · Uvicorn

Frontend: React 19 · Vite · TypeScript · Tailwind CSS · Framer Motion · TanStack Query

Databases & Infra: Supabase · PostgreSQL · SQLite · Redis (Upstash) · Vercel · Render · Git/GitHub · Linux/WSL

Specialized Domains: Combustion AI · Real-Time Telemetry · Sensor Data Fusion · Hyperspectral Imaging · Mental Health AI · Digital Twin Workflows

Tools: MATLAB · Jupyter · Figma · Plotly · Streamlit · Pandas · NumPy

KEY PROJECTS & RESEARCH

SENTINEL — Execution Intelligence Platform | AMD Slingshot Hackathon 2026 | Quantumstacks Lab *2026*

- Built end-to-end LLM-powered platform that extracts delivery commitments from Slack and Gmail, scores slip risk 0–100, and auto-alerts owners before deadlines are missed
- AI pipeline: FastAPI webhook ingestion → Groq LPU (Llama 3.3 70B) structured JSON extraction → 4-factor risk engine (deadline proximity, dependency chain, owner silence, workload)
- Interactive dependency graph visualization: React 19 + Framer Motion + TanStack Query; real-time Execution Health Score dashboard for leadership
- Full OAuth 2.0 integration: Slack, Gmail, GitHub; JWT auth; Supabase PostgreSQL with Row-Level Security; Celery + Upstash Redis for background task queuing
- Deployed: Vercel (frontend) + Render (backend); roadmap includes self-hosted Llama 3.3 70B on AMD Instinct MI300X via IBM Cloud

Stack: Python · FastAPI · React 19 · TypeScript · Tailwind CSS · Groq API · Llama 3.3 70B · Supabase · Redis · Vercel · Render

AI-Driven Combustion Instability Prediction — Hybrid Rocket Engines | Dr. M.G.R ACS Space Technology Centre *2025*

- TCN model: 92% accuracy for early warning of combustion instability; outperformed LSTM baselines and traditional statistical models
- Bayesian Optimization pipeline: 35% reduction in manual injector geometry experimentation iterations
- 8-channel sensor fusion (pressure, temperature, IMU) at 1000 Hz sampling rate; full digital twin simulation in MATLAB
- Presented at YASSC 2025 Regional & State levels; awarded 2nd Place and Certificate of Appreciation from State Evaluation Panel

Stack: Python · PyTorch · Scikit-optimize · NumPy · Pandas · MATLAB

DocuMind — Mental State Prediction using Deep Learning | Advanced Deep Learning Research *2024–2025*

- Stacked LSTM for time-series classification of mental health states (Stressed / Healthy / At_Risk) from 7-day social media and lifestyle metrics
- Achieved 81% Recall on At_Risk class through hyperparameter tuning and class weighting (28.74x for minority class); identified 5-day optimal prediction window

Stack: Python · TensorFlow/Keras · LSTM · Scikit-learn · Pandas

AIDEN AI — Intelligent Conversational Assistant | Space Club Initiative *2025*

- End-to-end conversational AI with NLP mood detection and RAG-style retrieval for adaptive academic guidance; dual recommendation engine achieving 88% user satisfaction

Stack: Python · HuggingFace Transformers · Flask API · SQLite

Ground Station Monitoring Software — Real-Time Telemetry Dashboard | CanSat & Rocket Avionics *2024–2025*

- Production-grade telemetry dashboard: 8+ sensor channels at 1000 Hz, <100ms latency via WebSocket architecture; pre-launch health monitoring + live anomaly detection
- Selected team for INSPACe Model Rocketry Competition national level

Stack: Python · Flask/FastAPI · WebSockets · Plotly · SQLite

Enterprise Data Automation & Analytics Pipeline | Personal Project *2025–2026*

- Fully automated Python pipeline that cleans messy CSV exports (date inconsistencies, financial formatting, duplicates) and delivers a real-time Streamlit BI dashboard
- Vectorized cleaning with Pandas + Regex sanitization; one-click upload to cleaned export with smart metrics and audit report

Stack: Python · Pandas · NumPy · Plotly Express · Streamlit

Smart Attendance System — Desktop Application | Personal Project *2025–2026*

- Full-featured desktop app with dual-mode architecture: Admin Dashboard (staff management, Excel export, DB backups) + Staff Terminal (PIN clock-in, auto late detection)
- Packaged as standalone Windows executable (.exe); SQLite local storage with zero server setup required

Stack: Python · CustomTkinter · SQLite · Pandas

CarbonCut — AI-Powered Waste Management | Smart India Hackathon 2025 *2024–2025*

- CNN image recognition: 94% accuracy across 6+ waste categories with real-time smartphone integration; selected for SIH 2025 national evaluation

Stack: Python · TensorFlow/Keras · React · Firebase · Google Maps API

PROFESSIONAL EXPERIENCE

Payload & Propellant Chemist / AI Research Engineer | Dr. M.G.R ACS Space Technology Centre, Chennai *2024 – Present*

- Led AI-driven combustion instability prediction research — TCN model achieving 92% accuracy deployed as real-time early warning system
- Conducted combustion behaviour analysis across 10+ oxidizer-fuel combinations; reduced chamber pressure prediction error by 18%
- Engineered end-to-end payload integration workflows for CubeSat and rocketry missions; 100% compliance with aerospace safety protocols
- Executed telemetry validation and sensor calibration; documented mission-critical findings supporting vehicle qualification and launch decisions

Mathematics & Chemistry Instructor | B.L Learning Institute, Chennai *2023 – 2024*

- Taught 50+ students; achieved 15–20% average performance improvement through personalized instruction and 40+ custom problem sets designed in Figma

EDUCATION

B.Tech — Computer Science & Engineering (Data Science & AI Specialization) | Dr. M.G.R Educational and Research Institute, Chennai *2023 – Present*

- GPA: 8.60/10.00 (Semester 4) | Upward trajectory: 7.38 → 7.71 → 8.14 → 8.60
- Relevant Coursework: Machine Learning · Deep Learning · Data Structures & Algorithms · Signal Processing · Computer Networks · Operating Systems

AWARDS, CERTIFICATIONS & LEADERSHIP

- AMD Slingshot Hackathon 2026 — SENTINEL Submission — Quantumstacks Lab *2026*
- 2nd Place — Paper Presentation (YASSC 2025) — Mechanical Engineering Department *Sep 2025*
- Certificate of Appreciation — Research Paper — State YASSC 2025 *Aug 2025*
- Runner-Up — Promptarhon AI Competition — AI: Beyond the Algorithm *May 2025*
- Selected Team — National Rocketry Competition — INSPACe Model Rocketry India *Oct 2025*
- Organising Committee Member — Space 4.0 National Seminar (ITCA) *2025*
- TiHAN Program Participant — IIT Hyderabad — Autonomous Navigation & Robotics *2024*

Certifications: State YASSC 2025 Research Paper · Promptarhon AI Workshop · Space 4.0 Organising Committee · TiHAN IIT Hyderabad · TCS LaunchPad C++ (2018)

LANGUAGES

English (Fluent) · Tamil (Native) · Hindi (Conversational) · German (Basic)